THE UNIVERSITY OF HONG KONG

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE

COMP3314 Machine Learning

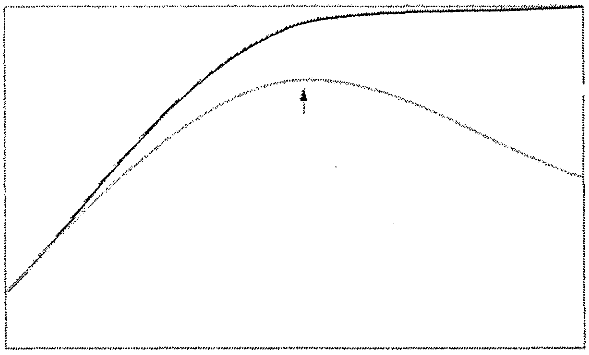
Date: May 15, 2018 **Time: 2:30 pm – 5:30 pm**

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

**** Answer ALL questions / Total marks is 100 ****

Question 1 [5%]

The following picture appears in one of the lecture notes. What is it meant to show? Please try to put the erased labels back in the picture: the two axes, the two curves, and areas/regions that have certain special significance.



Question 2 [10%]

The Naïve Bayes algorithm selects the best classification given a number of attribute values. Consider the following dataset from past cases of car thefts.

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

(a) Construct the likelihood table.

(b) Try using the Naïve Bayes algorithm to classify a "Red Domestic SUV". Show your steps. Looking at the probabilities, what can you comment about the result?

Question 3 [10%]

The XOR (Exclusive OR) function is well known for its being not linearly separable.

(a) Show that the Perceptron will not converge to a decision boundary (by trying a few rounds) for an XOR training dataset.

(b) Using an RBF kernel, the XOR dataset can be transformed into one that is linearly separable. Show how this is done.

Question 4 [10%]

(a) Explain the following equation (and all its component terms) that is used by Adaline in its learning process. How is it used during learning?

$$J(w) = \frac{1}{2}\sum_i \left(y^{(i)} - \phi\left(z^{(i)}\right)\right)^2$$

(b) Compare Adaline and Logistic Regression. What are their main differences? Why is Logistic Regression generally preferable in real practice?

Question 5 [10%]

"You could say that SVMs are successful because of one key insight, one neat trick." (Russell & Norvig, 2010)

(a) What is that key insight Russell and Norvig were referring to? Explain why that insight is so "key" to the success of SVMs.

(b) The neat trick refers to kernel trick. Explain why it would be dubbed as a "trick"? Use the example of a polynomial kernel to illustrate that.

Question 6 [10%]

Suppose there is this tiny Iris dataset with only 38 samples, of which 13, 16, and 9 are labelled Satosa, Versicolor, and Virginica respectively. A classifier predicted all the Satosas and Virginicas correctly, but only 10 of the Versicolors correctly (the others were classified as Virginica).

(a) Draw the confusion matrix for this result.

(b) Now suppose the classifier is converted to a binary one for classifying Versicolor and non-Versicolor. Draw the confusion matrix. What is the value of the precision? And of the recall? What are the literal meanings of these two values in this particular case here?

Question 7 [10%]

(a) Based on the following dataset that is related to tax refunds, create using the CART algorithm a decision tree for classifying whether a person would cheat or not. Show your steps, and draw the tree where nodes/leaves would display their essential information (just like the Iris example in the lecture notes). Assume no hyper-parameters have been set to constrain the growth of the tree.

(b) For the XOR example which we used to prove that the CART algorithm is not optimal (three binary features a, b, c; y = a XOR b; c == y 60% of the time), can you think of a particular dataset (with 10 samples) for which CART will produce the optimal tree? Show the dataset, and prove your case.

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Question 8 [10%]

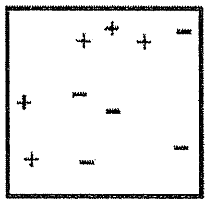
An ensemble classifier uses 3 base classifiers and simple majority voting for its classification task.

(a) If these base classifiers are "accurate", independent in their decision making, but have the same error rate, prove that the ensemble classifier has better error rate than a single base classifier.

(b) What about if the base classifiers in the above are not "accurate"?

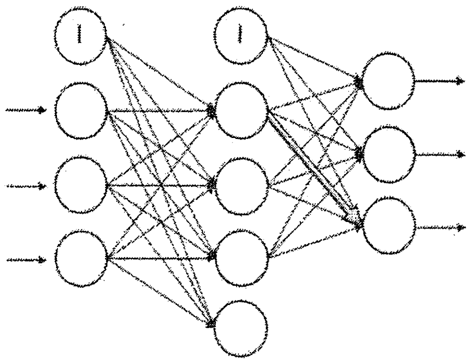
(c) Diversity is another desired property when combining multiple base classifiers. What are some of the ways of achieving diversity?

(d) Describe or depict the steps of AdaBoost: using several weak classifiers (e.g., decision tree) to arrive at a strong classifier for the dataset shown below.



Question 9 [10%]

(a) Given the MLP as shown below, and the various matrices that are involved including **X**, **A**, **Z**, **W**, and the activation function ϕ, (i) write out the equations that connect these various matrices and the activation function together in forming the forward feeding step of the MLP; use superscripts (in), (h), and (o) to mark and distinguish the matrices pertaining to the input, the hidden, and the output layer respectively; (ii) for each of the matrices that you have included in your answer to (i), write down its dimensions (i.e., "? x ?"), assuming that there are *n* samples in a mini-batch.



(b) Describe how backpropagation is carried out. The equations are optional, but your answer should include the terms: cost function, error (or error term or sigma), derivative, gradient, weight, eta (or the learning rate) and regularization.

(c) What are the advantages of dividing the training dataset into mini-batches (i.e., stochastic gradient descent)?

Question 10 [10%]

Use the k-means algorithm and Euclidean distance to cluster the following 8 data points into 3 clusters: A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9). The initial seeds (centroids) are A1, A4 and A7. Run the k-means algorithm for 1 iteration only. At the end of this iteration show:

(a) The new clusters (and their data points) and their centroids.

(b) How many more iterations you think will be needed to converge? Explain or show or prove.

Question 11 [5%]

(a) Contrast reinforcement learning with supervised and unsupervised learning in aspects including their goals, supervision, agent, environment, and training.

(b) Draw a small example of a Markov Decision Process, and use it to describe the various components that are in a typical Markov Decision Process.

(c) Use the example in (b) to explain what is Q-value.

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